

DIMUTHU WIJESIRIWARDENE

ROBOTICS & AUTONOMOUS SYSTEMS RESEARCHER

MOBILE ROBOTICS • PERCEPTION & CONTROL

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Location : Colombo, Sri Lanka

EDUCATION

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| ● BEng (Hons) Electrical and Electronic Engineering <ul style="list-style-type: none">○ Upper Second Class Honours | Curtin University
Perth, Australia | Jan 2016 - Jan 2020 |
| ● GCE Advanced Level <ul style="list-style-type: none">○ Physical Science Stream○ Math, Physics & Chemistry | Dharmapala Vidyalaya Pannipitiya
Colombo, Sri Lanka | Jan 2011 - Sep 2014 |

RESEARCH EXPERIENCE

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| ● Graduate Research Assistant | Faculty of Engineering University of Sri Jayewardenepura
Colombo, Sri Lanka | Oct 2021 - Jun 2022 |
| ● Undergraduate research assistant (Internship) | Sustainable Computing Research Lab (SCoRe Lab)
University of Colombo School of Computing
Colombo, Sri Lanka | Nov 2017 - Jan 2018 |

PROFESSIONAL EXPERIENCE

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| ● Application Engineer | Synopsys Inc. Colombo, Sri Lanka | Jun 2022 - Present |
| ● Associate Software Engineer | Insharp Technologies Pvt. Ltd. Colombo, Sri Lanka | Aug 2020 - Oct 2021 |

ROBOTICS AND RESEARCH PROJECTS

- Fire Rescue Assisting Robot ([Project Post](#))
 - Designed and developed an individual honors thesis investigating thermal perception for fire rescue robots
 - Developed a human detection system using a thermal sensor and computer vision techniques for smoke-filled environments for deployment on UGV/UAV robots
 - Final year individual research project at Curtin University (2019)
- NovasPlay - Omnidirectional Soccer robot ([Demo Video](#))
 - Designed and developed an omnidirectional soccer robot with a custom dribbling mechanism
 - Modeled and fabricated the robot using Autodesk Fusion, laser-cut sheet metal and 3D-printed components
 - Achieved a 5-1 victory in the initial round of the Synopsys robotics competition (2024)
- Reverse Vending Machine ([Demo Video](#))
 - Led the university side development of an industry collaborative automated Reverse Vending Machine project integrating a vision based identification system, embedded processing and automated sorting
 - Designed and simulated a 3D model of a reverse vending machine in Blender to aid prototype development
 - A research project at the University of Sri Jayewardenepura in collaboration with OREL Corporation (2022)
- Low-cost Time Domain Reflectometer ([Project Post](#))
 - Designed and built a low-cost time-domain reflectometer for electric fence fault detection as part of the elephant-human conflict mitigation research project
 - Developed and experimentally validated the prototype through circuit simulation, pulse generation, oscilloscope based measurements of signal reflections - Research project at SCoRe Lab (2017)
- Low-cost Fence Energizer ([Project Post](#))
 - Reverse engineered a GALLAGHER B180 Multi Power Fence Energizer through PCB trace analysis, continuity measurements and circuit reconstruction
 - Reconstructed the system in Multisim and supported the development of a lower-cost alternative as part of the elephant-human conflict mitigation research project at SCoRe Lab (2017)

- 3D Printed Quadruped Robot ([Demo Video](#))
 - Developed and fabricated a 3D-printed quadruped robot integrating servo actuators and embedded control hardware for research into mobility over challenging terrain
 - Assembled and validated the platform through actuator testing and locomotion experiments for a planned beach waste collection research project at SCoRe Lab (2018)
- The Wind Turbine ([Demo Video](#))
 - Designed, fabricated, assembled and evaluated a small-scale wind turbine prototype demonstrating mechanical to electrical energy conversion
 - Team member and presenter, Principles and Communication module project at the university (2016)
- The Automated Bedsheet ([Demo Video](#))
 - Led the development from initial concept through mechanical design, fabrication and functional prototyping of a system for automated bedsheet deployment and wrinkle reduction
 - Designed and fabricated a custom stepper motor driver reducing prototype hardware cost by approximately 85%
 - Presented the project at the university engineering exhibition (“Young Engineering Expo”, 2016)

ENGINEERING AND SOFTWARE PROJECTS

- Verdi - Debug and Verification Management Platform
 - Performed incoming bug analysis, root cause analysis and validation of new features and enhancements for customers such as Apple, Intel, NVIDIA, AMD
 - Developed test automation solutions using scripting and AI-Driven testing methodologies; contributed new regression tests to the regression suite, improving verification coverage and quality assurance processes
 - Served as the product validation owner of Verdi Transaction Debug component with quality assurance & signoff responsibilities
 - Worked with Synopsys VIP designs & UVM, performing tool specific testing - Project at Synopsys
- VCS - Verilog Compiled Simulator
 - Validated and debugged VCS Intelligent Coverage Optimization (ICO), an AI/ML driven technology for accelerating functional coverage closure in semiconductor verification
 - Validated integration of ICO with Verdi RDA to support identification of faulty code & traces - Project at Synopsys
- Euclide - Real-Time Hardware Development and Verification IDE
 - Validated new features & enhancements & performed RCA of customer reported issues - Project at Synopsys
- jTerm - Efficient Jira creation tool
 - Developed a tool for streamlined Jira creation with built-in support for screenshots and annotations via the in-house *Snipper* tool
 - Adopted by Synopsys engineering teams globally with 500+ Jiras created to date
 - Presented at the internal poster presentation event (“Purple Poster Event”, 2025) at Synopsys
- jCrash - Streamlined crash reporting tool
 - Developed a command line tool to report application crashes instantly by executing a single command
 - Received the Synopsys Innovations Award & Presented at the event (“Purple Poster Event”, 2024) at Synopsys
- gCron - Cronjob manager
 - Developed an intuitive interface for cron job management with natural-language scheduling, real-time previews and backup support; adopted by teams globally
 - Presented at the internal poster presentation event (“Purple Poster Event”, 2026) at Synopsys

- Additional Engineering and Software Projects
 - weGen - Weekly Report Generator
 - Unified project status & weekly reporting into a single interface
 - PerfTest - Automated Performance Testing
 - Automated Performance testing across multiple customer designs
 - Cdiff - Customized Diff Tool
 - Developed a customized diff utility for efficient engineering workflows
 - TuTi - Virtual Classroom
 - Developed a full stack virtual classroom platform at Inshar Technologies
 - Retailz - Data Analytics Platform
 - Developed a full stack data analytics platform for supermarkets and suppliers

PUBLICATIONS

- Wijesiriwardene, D. and Thilakumara, R. "Human Detection for Fire Rescue Robots Using a Thermal Sensor and Hu Moment Invariants", Manuscript submitted to *IEEE International Conference on Electrical Engineering and Emerging Technologies (ICEET)*, 2026.

AWARDS, PRESENTATIONS AND RECOGNITIONS

- 2024 - Award Recipient, Synopsys Innovations Award (*jCrash*)
- 2026 - Presenter, Purple Poster Event (*gCron*)
- 2025 - Presenter, Purple Poster Event (*jTerm*)
- 2024 - Presenter, Purple Poster Event (*jCrash*)
- 2024 - Competitor, Synopsys RoboCup Competition (*NovasPlay*) - [Demo Video](#)
- 2016 - Presenter, Young Engineering Expo (*Automated Bedsheet*) - [Demo Video](#)

TECHNICAL SKILLS

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| <ul style="list-style-type: none"> ● Robotics & Embedded Systems <ul style="list-style-type: none"> ○ Raspberry Pi, Arduino, PIC, MSP430, PLC ○ Sensors, Actuators ○ Robot Operating System (ROS 2) ○ Hardware Interfacing ○ I2C, SPI, UART, PWM | <ul style="list-style-type: none"> ● Programming & Software <ul style="list-style-type: none"> ○ Python, C, Shell scripting (Csh, Bash), Tcl ○ Version Control (Git) ○ Linux (Ubuntu, Raspbian) ○ AI-assisted Development ○ Prompt Engineering |
| <ul style="list-style-type: none"> ● Computer Vision, Simulation & Numerical Tools <ul style="list-style-type: none"> ○ OpenCV, Pandas ○ NumPy, SciPy ○ MATLAB ○ Godot Engine | <ul style="list-style-type: none"> ● CAD, Design & Fabrication <ul style="list-style-type: none"> ○ Autodesk Fusion, Blender, AutoCAD, SolidWorks ○ 3D-Printing (PLA, ABS, TPU, TPE), Sheet Metal Fabrication (laser cutting) ○ Mechanical Assembly; Power Tools |
| <ul style="list-style-type: none"> ● Electronics & PCB Design <ul style="list-style-type: none"> ○ Circuit Simulation: NI Multisim, Proteus ○ PCB Design: Eagle PCB ○ PCB Manufacturing: Toner-transfer Method ○ PCB Reverse Engineering ○ Oscilloscopes, Multimeters, Soldering | <ul style="list-style-type: none"> ● Validation, Verification & Quality Assurance <ul style="list-style-type: none"> ○ Synopsys Verdi (Debugging) ○ Synopsys VCS (Simulation) ○ Root Cause Analysis, Jira ○ Selenium, Test Planning ○ AI-based Test Case Creation Agents |
| <ul style="list-style-type: none"> ● Web & Software Development <ul style="list-style-type: none"> ○ Django, Laravel ○ HTML, CSS ○ JavaScript, Vue.js ○ Blockchain (Smart contracts, Solidity) | <ul style="list-style-type: none"> ● Technical Communication & Media <ul style="list-style-type: none"> ○ Poster Presentations ○ Technical Demonstrations ○ Video Production ○ Video Editing (Adobe Premiere Pro) |